Proof supplement: weighted federated synthetic data

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This post-experiment supplement preserves the report's theoretical framework. It does not claim originality for standard results, modify the frozen method, or certify its empirical outcomes. The fallback in Statement 8 is a theoretical proposal, not the frozen selection method.

Let K be the number of clients and $\Delta_K = \{w \geq 0 : \sum_i w_i = 1\}$. Let $\alpha \in \Delta_K$ denote target size weights and $\mathcal{W} \subseteq \Delta_K$ a nonempty closed feasible set containing α unless stated otherwise. Let Q_i be fixed fitted distributions, $q_1, \ldots, q_L$ fixed measurable queries taking values in $[0,1]$, $H_{\ell i} = \mathbb{E}_{Q_i} q_\ell$, and $Q_w = \sum_i w_i Q_i$. Let $B_{\ell i}$ denote the empirical answer on reference client i and $b_R = B\alpha$. For a generated table D_S, let b_S be its query vector, $\mathcal{L}_{\mathcal{Q}}(D_S, D_R; \alpha) = \|b_S - b_R\|_\infty$, and $L_R(w) = \|Hw - b_R\|_\infty$. The latter compares *model expectations* to a fixed finite real reference, whereas $\mathcal{L}_{\mathcal{Q}}$ compares a realized finite synthetic table to that reference. Sampling and allocation error connect them; neither objective is automatically population loss.

For measures A, B, $d_{\mathcal{Q}}(A,B) = \max_{q \in \mathcal{Q}} |\mathbb{E}_A q - \mathbb{E}_B q|$ and $\mathrm{TV}(A,B) = \sup_E |A(E) - B(E)|$. Write $\rho_m(w) = \frac{1}{2}\|\bar{w} - w\|_1$ for realized allocation proportions $\bar{w}$. For a pilot matrix $\widehat{H}$ and target summary $\widehat{b}$, assume the error event

$$\max_i \left\| \widehat{H}_{\cdot i} - H_{\cdot i} \right\|_\infty \leq r_H, \qquad \left\| \widehat{b} - b_R \right\|_\infty \leq r_b, \quad r = r_H + r_b. \tag{1}$$

Set $\widehat{L}(w) = \left\| \widehat{H}w - \widehat{b} \right\|_\infty$. For prediction, $R_A(g) = \mathbb{E}_A \ell(g(X), Y)$ with $0 \leq \ell \leq 1$, and $d_{\ell,\mathcal{G}}(A,B) = \sup_{g \in \mathcal{G}} |R_A(g) - R_B(g)|$. In the Brier statement, R_A specifically uses squared loss. Conditional notation R_C, T_C means conditioning on C with positive mass. All optimization tolerances are nonnegative; all logarithms are natural, $\lambda \geq 0$ where used as a penalty, and $\varepsilon \geq 0$ in the transport statement.

For hypothetical population laws P_i, define $e_i = \mathrm{TV}(P_i, Q_i)$, $D_Q = \max_{i,j} \mathrm{TV}(Q_i, Q_j)$, and $a_A = \mathrm{TV}(Q_w, A_w)$. Inserting Q_α and Q_w and using mixture convexity and the positive/negative weight decomposition yields $\mathrm{TV}(\sum_i \alpha_i P_i, A_w) \leq \sum_i \alpha_i e_i + (D_Q/2)\|w - \alpha\|_1 + a_A$. These population quantities were not certified by the finite-query experiment.

The achievable model-answer set is $H\Delta_K = \mathrm{conv}\{H_{\cdot i}\}$, so $\inf_{w \in \Delta_K} L_R(w) = \mathrm{dist}_\infty(b_R, H\Delta_K)$. For $Q_w(C) > 0$, conditioning routes via weights $w_i Q_i(C)/Q_w(C)$; zero-mass components contribute zero. A finite exact joint model has posterior $\eta_{Q_w}(x) = \sum_i w_i q_i(x,1) / \sum_i w_i q_i^X(x)$ when the denominator is positive. These identities do not imply a learned adaptive gate.

Statement 1 (Sharp representation bound; thm:representation)**.** *For $H \in [0,1]^{L \times K}$ and $w, \alpha \in \Delta_K$, let $D_H = \max_{i,j} \|H_{\cdot i} - H_{\cdot j}\|_\infty$. Then*

$$\|H(w - \alpha)\|_\infty \leq \frac{D_H}{2}\|w - \alpha\|_1 \leq \frac{1}{2}\|w - \alpha\|_1 . \tag{2}$$

The constant $1/2$ is attained by some two-client query matrices.

Proof. Let $v = w - \alpha$ and $s = \sum_{v_i>0} v_i = \frac{1}{2}\|v\|_1$. For $s > 0$, its positive and negative parts, divided by s, are probability vectors. A row h therefore satisfies $|hv| \le s(\max_i h_i - \min_i h_i)$. Maximize over rows. For sharpness, take $K = 2$, $h = (0, 1)$, $\alpha = (1, 0)$, and $w = (1 - t, t)$. Both sides of the last inequality equal t. For a fixed row, equality requires its extrema on the respective positive and negative supports of v. □

Statement 2 (Observable error decomposition; thm:decomp)**.** *Let $B_{\ell i}$ be the empirical query mean in D_i^R, so $b_R = B\alpha$. Then*

$$\|b_S - B\alpha\|_\infty \le \|b_S - Hw\|_\infty + \sum_i \alpha_i \|H_{\cdot i} - B_{\cdot i}\|_\infty + \frac{D_H}{2}\|w - \alpha\|_1 . \tag{3}$$

For an additional aggregation operator or compressed model, add the discrepancy between its query-answer vector and Hw.

Proof. The exact vector identity is $b_S - B\alpha = (b_S - Hw) + (H - B)\alpha + H(w - \alpha)$. Apply the triangle inequality, convexity, and Statement 1. An operator approximation is another inserted vector. The components need not be independent, positive contributions: errors may cancel before norms are taken. □

Statement 3 (Subset collapse and identifiability; thm:subsets)**.** *If a subset generator is $Q_B = \sum_{i\in B} w_i^{(B)} Q_i$ and subsets are mixed with weights ρ_B, then*

$$\sum_B \rho_B Q_B = \sum_i \omega_i Q_i, \qquad \omega_i = \sum_{B\ni i} \rho_B w_i^{(B)}. \tag{4}$$

Regrouping introduces no new mixture distributions. Query answers identify all simplex weights uniquely if $\ker H \cap \{v : \mathbf{1}^\top v = 0\} = \{0\}$*; otherwise feasible null directions can make different weights observationally indistinguishable.*

Proof. Exchange the finite sums; nonnegativity and normalization give $\omega \in \Delta_K$. Equality of two answer vectors implies $H(w - w') = 0$ and $\mathbf{1}^\top(w - w') = 0$. The kernel condition is sufficient and necessary for injectivity on the whole simplex because any nonzero zero-sum direction can be scaled around an interior point. Boundary constraints can yield uniqueness at a particular point without global injectivity. □

Statement 4 (Model-only information is insufficient; ex:ident)**.** *Suppose identical releases in two possible worlds are $Q_1 = \text{Bernoulli}(1/4)$ and $Q_2 = \text{Bernoulli}(3/4)$. No assumption connects model error to the real reference. For $q(z) = z$, a real empirical target with fraction $1/4$ of ones has oracle $w = (1, 0)$, while fraction $3/4$ has oracle $w = (0, 1)$. A release-only rule returns the same answer x in both worlds. At least one error is $\max\{|x - 1/4|, |x - 3/4|\} \ge 1/4$. Randomization cannot remove this worst-world expected lower bound, since the two absolute errors sum to at least $1/2$ pointwise. Informative summaries or modelling assumptions are necessary.*

Statement 5 (Uniform perturbation and oracle inequality; thm:oracle)**.** *Uniformly over every $w \in \Delta_K$,*

$$|\widehat{L}(w) - L_R(w)| \le r. \tag{5}$$

Let $\Omega(w) = \frac{1}{2}\|w - \alpha\|_2^2$ and let $\widehat{w}_\lambda \in \mathcal{W}$ minimize $\widehat{L}(w) + \lambda\Omega(w)$ to tolerance $\zeta \ge 0$. If $w^* \in \arg\min_{w\in\mathcal{W}} L_R(w)$, then

$$L_R(\widehat{w}_\lambda) - L_R(w^*) \le 2r + \lambda\Omega(w^*) + \zeta \le 2r + \lambda + \zeta. \tag{6}$$

This is valid for data-adaptive weights chosen from the same finite pilot.

Proof. For simplex w, $\left\|(\widehat{H}-H)w\right\|_\infty \le r_H$. The reverse triangle inequality gives (5). Compare the approximate regularized minimizer with w^*, use (5) twice, and drop $-\lambda\Omega(\widehat{w}_\lambda) \le 0$. Since two simplex points have squared distance at most two, $\Omega \le 1$. The argument is uniform in w and therefore covers selection of weights from $\widehat{H}, \widehat{b}$; it does not cover arbitrary data-adaptive query discovery without an additional uniform event. □

Statement 6 (Regularization stability; thm:stability)**.** *For two objectives F, F' that share the term $\lambda\Omega$, $\lambda > 0$, suppose their remaining terms are convex on the same convex feasible set and $\sup_w |F(w) - F'(w)| \le \eta$. Their unique exact minimizers w, w' obey*

$$\|w - w'\|_2 \le 2\sqrt{\eta/\lambda}. \tag{7}$$

Proof. Strong convexity and constrained optimality give $\frac{\lambda}{2}\|w - w'\|_2^2 \le F(w') - F(w)$. Adding and subtracting F' and using optimality of w' bounds the right side by 2η. Strong convexity also gives uniqueness. Stability increases with λ, at the cost of the comparator term in (6). □

Statement 7 (Finite-workload stratified sampling; thm:sampling)**.** *Condition on fitted models and selected weights. Generate m_i independent rows from Q_i, with $\sum_i m_i = m$, and write $\bar{w}_i = m_i/m$. For fixed L bounded queries, with probability at least $1-\gamma$,*

$$\|b_S - Hw\|_\infty \le u_m(\gamma) + \rho_m(w), \quad u_m(\gamma) = \sqrt{\frac{\log(2L/\gamma)}{2m}}, \tag{8}$$

where $\rho_m(w) = \frac{1}{2}\|\bar{w} - w\|_1$. Largest-remainder allocation gives $\rho_m(w) < K/(2m)$. The same u_m applies to IID mixture sampling with $\rho_m = 0$.

Proof. Each query average is a mean of m independent variables in $[0,1]$, not necessarily identically distributed. Its expectation is $H\bar{w}$. Hoeffding's exponential argument yields tail probability at most $2e^{-2mu^2}$ per query. A union bound gives the displayed u_m. Statement 1 bounds $\|H(\bar{w} - w)\|_\infty$ by ρ_m. Each rounded count differs from mw_i by less than one. Shuffling does not change query averages. □

Statement 8 (Sufficient fallback certificate; thm:fallback)**.** *On the error event* (1)*, choose any candidate $\widehat{w} \in \mathcal{W}$ using the released pilot and summaries. Let $\kappa > 0$ be a desired gain. Accept only when*

$$\widehat{L}(\alpha) - \widehat{L}(\widehat{w}) \ge \kappa + 2r + 2u_m(\gamma_S/2) + \rho_m(\alpha) + \rho_m(\widehat{w}). \tag{9}$$

Generate fresh candidate and size tables independently of selection. With conditional probability at least $1-\gamma_S$, acceptance guarantees

$$\mathcal{L}_\mathcal{Q}(D_S^{\widehat{w}}, D_R; \alpha) \le \mathcal{L}_\mathcal{Q}(D_S^{\alpha}, D_R; \alpha) - \kappa. \tag{10}$$

On rejection, returning the already generated size table has exactly its loss. If the error event (1) *holds with probability $1-\gamma_0$, the joint guarantee is at least $1-\gamma_0-\gamma_S$.*

Proof. Uniform perturbation gives $L_R(\widehat{w}) - L_R(\alpha) \le \widehat{L}(\widehat{w}) - \widehat{L}(\alpha) + 2r$. Apply Statement 7 to the two fresh tables and use a union bound. The sum of their deviations is the last three terms in (9). Returning the same baseline table on rejection avoids falsely promising equality between two independent baseline draws. □

Statement 9 (Empirical conditional transfer; thm:conditional)**.** *For two fixed measures R, S with $r = R(C) > 0$, $s = S(C) > 0$, let $e_C = \max_{q \in \mathcal{Q}_C} |\mathbb{E}_S[\mathbf{1}_C q] - \mathbb{E}_R[\mathbf{1}_C q]|$ and $e_0 = |s - r|$. Then*

$$d_{\mathcal{Q}_C}(S(\cdot \mid C), R(\cdot \mid C)) \le \min\{1, (e_C + e_0)/r\}. \tag{11}$$

Proof. For $a = \mathbb{E}_R[\mathbf{1}_C q]$ and $c = \mathbb{E}_S[\mathbf{1}_C q]$, add and subtract c/r in $|c/s - a/r|$. The result is at most $|c - a|/r + c|r - s|/(rs) \le (|c - a| + |r - s|)/r$, since $c \le s$. Take the maximum and cap by one. If $s = 0$, the synthetic conditional is undefined and must abstain. □

Statement 10 (Matching marginals can reverse a conditional; ex:parity)**.** *Let R put mass $1/2$ at $(X, Y) = (0, 0), (1, 1)$ and S at $(0, 1), (1, 0)$. Their one-way means agree exactly. Nevertheless $R(Y = 1 \mid X = 1) = 1$ and $S(Y = 1 \mid X = 1) = 0$. A perfect synthetic predictor $g(x) = 1 - x$ has real squared loss one. Thus neither conditional fidelity nor predictive utility follows from these marginal matches. Higher-order parity distributions give analogous failures for workloads omitting the relevant interaction.*

Statement 11 (Simultaneous pilot accuracy; thm:pilot)**.** *Conditional on fitted models and a fixed workload, suppose each column of $\widehat{H}$ averages p_i independent draws from Q_i. With probability at least $1 - \gamma_H$,*

$$\left\| \widehat{H}_{\cdot i} - H_{\cdot i} \right\|_\infty \le \sqrt{\frac{\log(2KL/\gamma_H)}{2p_i}} \quad \textit{for every } i. \tag{12}$$

The maximum of these client-specific radii is a valid r_H in Statement 5.

Proof. For each client/query pair Hoeffding's bound is $2\exp(-2p_i u_i^2)$. Set each to $\gamma_H/(KL)$ and sum over KL events. Independence between different queries is unnecessary. Fresh within-client sampling is sufficient; data-derived models and queries can be conditioned upon if pilot draws are independent of their construction. □

Statement 12 (A sufficient target-summary radius; thm:summary)**.** *Fix α and $\mathcal{Q}$. For each i, suppose calibration and reference observations are independent, bounded-query evaluations of sizes $c_i, t_i > 0$, drawn from the same client law; observations are independent within and across these samples. Let b_{cal} and b_R use the same α. Then with probability at least $1 - \gamma_b$,*

$$\|b_{\text{cal}} - b_R\|_\infty \le r_{\text{data}} = \sqrt{\frac{\log(2L/\gamma_b)}{2} \sum_i \alpha_i^2 \left(\frac{1}{c_i} + \frac{1}{t_i} \right)}. \tag{13}$$

If $\widehat{b} = b_{\text{cal}} + e$ and $\|e\|_\infty \le r_{\text{noise}}$, then $r_b = r_{\text{data}} + r_{\text{noise}}$ is valid. For exact released summaries and a calibration target itself, $r_b = 0$; this does not make that target identical to a held-out reference.

Proof. For each query, the calibration-minus-reference mean is a sum of independent mean-zero bounded variables. Their interval lengths are α_i/c_i repeated c_i times and α_i/t_i repeated t_i times. The sum of squared lengths is $V = \sum_i \alpha_i^2 (1/c_i + 1/t_i)$. Hoeffding gives $2\exp(-2r^2/V)$; a union bound yields (13). Add the release error by the triangle inequality. □

Statement 13 (Validation-to-reference comparison; thm:val)**.** *For the same model-answer map H, suppose $\|b_{\text{val}} - b_R\|_\infty \le r_V$. If w_V minimizes $\|Hw - b_{\text{val}}\|_\infty$ over $\mathcal{W}$ to tolerance ζ, then*

$$L_R(w_V) \le \inf_{w \in \mathcal{W}} L_R(w) + 2r_V + \zeta. \tag{14}$$

If H is replaced by a pilot matrix, add $2r_H$.

Proof. The reverse triangle inequality bounds the difference of the two objectives by r_V for every w. Apply this at w_V and at a comparator minimizing L_R, with the optimization inequality between them. The pilot statement follows from Statement 5. □

Statement 14 (Projected probability and information loss; thm:projection)**.** *Let $T : \mathcal{Z} \to \mathcal{Z}$ be deterministic. On a finite or countable support,*

$$(T_{\#}Q)(z) = \sum_{u:T(u)=z} Q(u). \tag{15}$$

This is normalized and satisfies $\mathbb{E}_{T_{\#}Q} q = \mathbb{E}_Q(q \circ T)$. For arbitrary measures A, B, $\mathrm{TV}(T_{\#}A, T_{\#}B) \leq \mathrm{TV}(A, B)$.

Proof. The preimages partition the original support, so their masses sum to one. The expectation identity follows by grouping terms by image. For any event E, $(T_{\#}A)(E) - (T_{\#}B)(E) = A(T^{-1}E) - B(T^{-1}E)$, bounded by the original TV. Take the supremum. □

Statement 15 (Support-pilot statement and rejection cost; thm:support)**.** *For a fixed condition C and p independent IID proposals from a model Q, let $\widehat{c}$ be the matching fraction and $c = Q(C)$. With probability $1 - \gamma$,*

$$c \geq \widehat{c} - \sqrt{\log(1/\gamma)/(2p)}. \tag{16}$$

For a largest-remainder pilot from mixture Q_w, subtract an additional $\rho_p(w)$ to obtain a bound on $Q_w(C)$. For J predeclared conditions, replace γ by γ/J. Under IID rejection with $c > 0$, the expected proposals for a accepted rows is a/c, and the probability of failing a cap B is $\mathbb{P}\{\mathrm{Binomial}(B, c) < a\}$.

Proof. Apply the one-sided Hoeffding inequality to indicators. With fixed client counts, indicators are independent but may have different means; their average expectation is $Q_{\bar{w}}(C)$, within $\rho_p(w)$ of $Q_w(C)$. For IID proposals, waiting times between accepts are geometric with mean $1/c$; their sum has mean a/c, and accepted counts before the cap are binomial. □

Statement 16 (Student filtering and transfer; thm:student)**.** *Let M be a raw per-condition student distribution with $a = M(C) > 0$, and let $M_C = M(\cdot \mid C)$. Then $\mathrm{TV}(M, M_C) = 1 - a$. For any fixed conditional workload and teacher conditional T_C,*

$$d_{\mathcal{Q}_C}(M_C, R_C) \leq d_{\mathcal{Q}_C}(M_C, T_C) + d_{\mathcal{Q}_C}(T_C, R_C). \tag{17}$$

An average error over conditions does not bound a worst-case error without further support or complexity restrictions.

Proof. Write $M = aM_C + (1 - a)M_{C^c}$, with disjoint conditional supports; the event C^c attains TV $1 - a$. The transfer inequality is the querywise triangle inequality followed by a maximum. For the last statement, place an error of one on a condition of evaluation probability η and zero elsewhere. The average is η while the maximum is one, for any positive η. □

Statement 17 (Abstract TV conditional implication; thm:tvconditional)**.** *For arbitrary measures A, B, suppose $\mathrm{TV}(A, B) \leq e$ and $A(C) \geq \tau > e$. Then $B(C) > 0$ and*

$$\mathrm{TV}(A(\cdot \mid C), B(\cdot \mid C)) \leq \min\{1, 2e/\tau\}. \tag{18}$$

A separate conditional student with TV error a_C from $B(\cdot \mid C)$ adds a_C to this upper bound, capped at one.

Proof. $B(C) \geq A(C) - e > 0$. Apply the proof of Statement 9 to every indicator of an event inside C: both joint-event and mass discrepancies are at most e. Take the event supremum and then the triangle inequality for the student. □

Statement 18 (Risk transfer and approximate empirical learning; thm:risk)**.** *Suppose $d_{\ell,\mathcal{G}}(R_\alpha, Q_w) \leq e_\ell$ and a synthetic sample satisfies $\sup_{g\in\mathcal{G}} |\widehat{R}_{D_S}(g) - R_{Q_w}(g)| \leq \xi_m$. If g_S is an approximate synthetic empirical-risk minimizer with tolerance ζ, then*

$$R_{R_\alpha}(g_S) - \inf_{g\in\mathcal{G}} R_{R_\alpha}(g) \leq 2e_\ell + 2\xi_m + \zeta. \tag{19}$$

For arbitrary measures A, B, $d_{\ell,\mathcal{G}}(A, B) \leq \mathrm{TV}(A, B)$. For finite fixed $|\mathcal{G}| = M$ and IID synthetic draws, one sufficient radius is $\xi_m = \sqrt{\log(2M/\gamma)/(2m)}$ with probability $1 - \gamma$.

Proof. For any comparator g, move from real risk of g_S to model risk, then empirical synthetic risk, use approximate empirical optimality, and reverse the two comparisons for g. This incurs two copies of e_ℓ and ξ_m, plus ζ; take the infimum. The TV implication follows by integrating the layer-cake indicators of a $[0, 1]$-valued loss. The finite-class statement is Hoeffding plus a union bound. Stratified generation adds the allocation correction as in Statement 7. □

Statement 19 (Brier decomposition and teacher consistency; thm:brier)**.** *For binary Y, let $\eta_A(x) = A(Y = 1 \mid X = x)$. For any $g : \mathcal{X} \to [0, 1]$,*

$$R_A(g) - R_A(\eta_A) = \mathbb{E}_{A^X}(g - \eta_A)^2. \tag{20}$$

If $\mathrm{TV}(A, B) \leq e$ and $a_g = \mathbb{E}_{B^X}(g - \eta_B)^2$, then

$$\mathbb{E}_{A^X}(g - \eta_A)^2 \leq a_g + 2e. \tag{21}$$

Proof. Condition on X: $\mathbb{E}[(g - Y)^2 \mid X] = (g - \eta_A)^2 + \eta_A(1 - \eta_A)$. Subtract the Bayes risk and integrate. For transfer, $R_A(g) \leq R_B(g) + e = R_B(\eta_B) + a_g + e \leq R_B(\eta_A) + a_g + e \leq R_A(\eta_A) + a_g + 2e$. □

Statement 20 (Restriction and smoothing on a finite support; thm:finite)**.** *Let a discrete target model Q assign positive mass to finite candidate support $\mathcal{S}$, and let $Q_\mathcal{S} = Q(\cdot \mid \mathcal{S})$. Then*

$$\mathrm{TV}(Q, Q_\mathcal{S}) = Q(\mathcal{S}^c). \tag{22}$$

If $\widehat{Q}_\mathcal{S}$ has normalized mass-vector error at most η in ℓ_1, and smoothing forms $Q' = (1 - \beta)\widehat{Q}_\mathcal{S} + \beta U$, then

$$d_\mathcal{Q}(Q, Q') \leq \min\{1, Q(\mathcal{S}^c) + \eta/2 + \beta\}. \tag{23}$$

Applying the same deterministic projection to both sides does not enlarge the TV version of this bound.

Proof. Restriction is the filtering identity of Statement 16. On a discrete support TV is half the ℓ_1 mass difference. Mixing in βU changes TV by at most β. Sum these quantities, use $d_\mathcal{Q} \leq \mathrm{TV}$, and apply Statement 14. □

Statement 21 (Geometric-score perturbation; thm:logpool)**.** *On common finite support, let positive unnormalized masses be $g(z) = \exp h(z)$ and $g'(z) = \exp h'(z)$, with $\max_z |h'(z) - h(z)| \leq \eta$. Normalize them to G, G'. Then*

$$\begin{aligned} e^{-2\eta} \leq G'(z)/G(z) \leq e^{2\eta}, \\ d_\mathcal{Q}(G, G') \leq \min\{1, (e^{2\eta} - 1)/2\}. \end{aligned} \tag{24}$$

For a fixed geometric weight vector, client log-score perturbations η_i give $\eta \leq \sum_i w_i\eta_i$.

Proof. The pointwise unnormalized ratio lies in $[e^{-\eta}, e^{\eta}]$, as does the ratio of normalizing constants. Dividing gives the first bound. Sum $\frac{1}{2}G(z)|G'(z)/G(z) - 1|$ and bound the absolute ratio deviation by $e^{2\eta} - 1$. The weighted score statement is the triangle inequality. □

Statement 22 (Finite entropic transport objective boundary; thm:ot)**.** *On finite probability supports with at most B points each, define*

$$T_\varepsilon(a, b) = \min_{\pi \in \Pi(a,b)} \langle C, \pi \rangle + \varepsilon \, \mathrm{KL}(\pi \| a \otimes b), \tag{25}$$

and T_0 without the penalty. Then $T_0 \le T_\varepsilon \le T_0 + \varepsilon \log B$. For fixed client weights, if a_ε minimizes $F_\varepsilon(a) = \sum_i w_i T_\varepsilon(a, b_i)$ to tolerance ζ over a common allowed set $\mathcal{A}$, then

$$F_0(a_\varepsilon) - \inf_{a \in \mathcal{A}} F_0(a) \le \varepsilon \log B + \zeta. \tag{26}$$

A cost perturbation bounded entrywise by κ adds 2κ when comparing the perturbed minimizer to the original objective. Restricting $\mathcal{A}$ further adds its approximation gap relative to a larger decision set.

Proof. For a coupling, $\mathrm{KL}(\pi \| a \otimes b)$ is mutual information and lies between zero and $\min\{H(a), H(b)\} \le \log B$. Evaluate the penalized objective at an unpenalized optimal coupling for the upper bound; nonnegativity gives the lower bound. Hence $F_0 \le F_\varepsilon \le F_0 + \varepsilon \log B$ uniformly, which yields (26) by comparison at its minimizer. A unit-mass coupling changes its cost by at most κ, giving two copies through a minimizer comparison. A restricted minimum differs from the larger-set minimum by exactly the stated nonnegative approximation gap. □

Statement 23 (Scope of composed release guarantees; thm:privacy)**.** *Suppose each full client mechanism is $(\varepsilon_i, \delta_i)$-DP, each client mechanism depends only on its own dataset and uses randomness independent across clients (conditional on the fixed input datasets), membership is fixed and disjoint, and record replacement affects only one client. Joint release is $(\max_i \varepsilon_i, \max_i \delta_i)$-DP. Subsequent server operations using only that release and independent randomness preserve the guarantee. If multiple mechanisms access overlapping records with the required conditional DP guarantees, basic sequential composition instead sums their ε and δ budgets. These conclusions are conditional on valid guarantees for every released mechanism; reusing hidden random noise is not justified by marginal per-fit guarantees alone.*

Proof. Under fixed disjoint membership, a neighboring record replacement changes only one client output law. Condition on the unaffected outputs and apply that client's DP inequality; integrate and take the largest budgets. Server postprocessing replaces an event by a randomized event probability in $[0, 1]$ and integrates the same inequality. For overlapping accesses, the standard adaptive composition theorem supplies the sum of budgets; disjoint parallel reasoning is inapplicable. The displayed theorem is an application of these established rules, not a new composition result. □

Statement 24 (Exact-model size-weight optimality; thm:sizeexact)**.** *Let $\alpha \in \Delta_K$, let $P_R = \sum_i \alpha_i P_i$ be the declared target and suppose the released generators satisfy $Q_i = P_i$ for every client. Then $w = \alpha$ globally minimizes both $\mathrm{KL}(P_R \| Q_w)$ and $d_{\mathcal{Q}}(P_R, Q_w)$, with minimum zero, where $Q_w = \sum_i w_i Q_i$. The KL minimizer is unique if the fixed component measures are affinely independent. Query-loss uniqueness requires affine independence of their query-answer vectors.*

Proof. By the assumptions, $Q_\alpha = \sum_i \alpha_i Q_i = P_R$. Both discrepancies are nonnegative and vanish at equality. For KL, a second zero-loss mixture must equal P_R as a measure. If the only coefficients v satisfying $\sum_i v_i = 0$ and $\sum_i v_i P_i = 0$ are $v = 0$, its weights must be α. For a query matrix B, the corresponding condition is $\ker(B) \cap \{v : \mathbf{1}^\top v = 0\} = \{0\}$. □

For finite discrete models, expected negative log likelihood equals $H(P_R) + \mathrm{KL}(P_R \| Q_w)$ when well-defined, so the exact-model result is a population likelihood statement. A separate multinomial likelihood for independently sampled client labels is maximized by their empirical proportions; counts fixed by design do not establish that sampling model. Neither result asserts that α maximizes the observed-data mixture likelihood in a finite sample: for example, with exact Bernoulli components $Q_1 = \text{Bernoulli}(0.2)$, $Q_2 = \text{Bernoulli}(0.8)$ and $\alpha = (1/2, 1/2)$, a single observed one has mixture likelihood $0.2w_1 + 0.8w_2$, maximized at $w = (0, 1)$.

Statement 25 (Small local error can favour different weights; ex:sizefails)**.** *Let* $P_1 = \text{Bernoulli}(0.2)$, $P_2 = \text{Bernoulli}(0.8)$ *and* $\alpha = (0.5, 0.5)$. *Their target is* $\text{Bernoulli}(0.5)$. *Suppose* $Q_1 = \text{Bernoulli}(0.3)$ *and* $Q_2 = P_2$. *Size weighting gives probability 0.55 of one and error 0.05 for* $q(z) = z$. *Weights* $w = (0.6, 0.4)$ *give probability 0.5 and zero error. They satisfy* $\|w - \alpha\|_1 = 0.2 \leq 0.25$ *and* $w_i \geq 0.5\alpha_i$. *The same alternative achieves zero KL, while size weighting gives approximately 0.005025. Hence unrestricted optimality of size weights for imperfect models is false, even within the retained constraints.*

Statement 26 (Approximate size optimality and weight localization; thm:sizenear)**.** *Define* $e_i = \|H_{\cdot i} - B_{\cdot i}\|_\infty$, $e_\alpha = \sum_i \alpha_i e_i$, *and* $e_{\max} = \max_i e_i$. *Let* $\mathcal{W}$ *be a closed subset of the simplex containing* α, *and let* w^* *minimize* $L_R(w) = \|Hw - B\alpha\|_\infty$ *over* $\mathcal{W}$. *Then*

$$0 \leq L_R(\alpha) - L_R(w^*) \leq e_\alpha. \tag{27}$$

If, additionally, a constant $c > 0$ *satisfies*

$$\|Bv\|_\infty \geq c \|v\|_1 \quad \textit{whenever } \mathbf{1}^\top v = 0, \tag{28}$$

then

$$\|w^* - \alpha\|_1 \leq \frac{e_\alpha + e_{\max}}{c}. \tag{29}$$

Proof. Since $b_R = B\alpha$, convexity of the norm gives $L_R(\alpha) = \|(H - B)\alpha\|_\infty \leq e_\alpha$. The first bound follows from optimality and nonnegativity. For any simplex w,

$$L_R(w) = \|B(w - \alpha) + (H - B)w\|_\infty \tag{30}$$
$$\geq c \|w - \alpha\|_1 - e_{\max}. \tag{31}$$

Use $L_R(w^*) \leq L_R(\alpha) \leq e_\alpha$ and rearrange. □

Statement 27 (A sufficient query-loss certificate; thm:sizedual)**.** *Let* $r = H\alpha - b_R$ *and* $t = \|r\|_\infty > 0$. *Let* $\mathcal{A} = \{\ell : |r_\ell| = t\}$. *Suppose there are* $\lambda_\ell \geq 0$ *summing to one over* $\mathcal{A}$ *such that* $s_\ell = \lambda_\ell \operatorname{sign}(r_\ell)$ *for* $\ell \in \mathcal{A}$, $s_\ell = 0$ *otherwise, and* $H^\top s = \nu \mathbf{1}$ *for a scalar* ν. *Then* α *is a global minimizer of* $\|Hw - b_R\|_\infty$ *over the entire simplex, and therefore over any feasible subset containing it. If* $t = 0$, *optimality already follows from nonnegativity.*

Proof. We have $\|s\|_1 = 1$ and $s^\top r = t$. For any simplex vector w, norm duality and the zero-sum constraint give

$$\|Hw - b_R\|_\infty \geq s^\top (Hw - b_R) \tag{32}$$
$$= s^\top r + \nu \mathbf{1}^\top (w - \alpha) = t. \tag{33}$$

Size weights attain this lower bound. □